

ML Project 2024-25: Root Node Identification

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I. INTRODUCTION

This investigation presents the complete process to develop a model designed to address the problem of root word identification, referred to herein as **ROOT NODE IDENTIFICATION**. The objective is to accurately determine the root word in each sentence from a multilingual corpus encompassing 21 languages. The root word is defined, albeit loosely, as the most "important" word in a sentence. However, the notion of "importance" is intentionally left underspecified; it may relate to syntactic prominence, semantic content, lexical weight, contextual relevance, or some combination thereof.

The central hypothesis of this work is that graph-theoretic centrality scores are meaningfully related to the identification of root words. To investigate this, each sentence in the corpus is modeled as an undirected graph, where nodes represent words and edges connect words that share a syntactic relationship.

Importantly, we use undirected rather than directed graphs to prevent the model from leveraging explicit edge direction (e.g., in-degree or word order) as a shortcut. This design choice ensures that any insights the model gains about root identification are derived from centrality measures rather than from structural cues directly tied to root positioning.

A. Motivation and Context

This problem space remains largely unexplored in the literature, likely due to its highly abstract and synthetic nature. As such, this work does not aim to provide an exhaustive literature review but instead focuses on the experimental frameworks employed. We acknowledge the foundational contributions of Bishop [1] and Marta Arias' slides and notes about machine learning [2], whose insights on pattern recognition have informed several aspects of our modeling pipeline.

B. Structure of the Paper

The remainder of this paper is organized as follows:

- Section II provides a detailed analysis of the dataset and preprocessing steps.
- Section III outlines the modeling framework, including reproducibility practices, prediction strategy, and techniques for handling class imbalance.
- Section IV presents the experimental results and performance evaluations.
- Section V introduces the final and most effective formulation of the problem based on empirical findings.

- Section VI concludes the paper with a high-level summary of contributions, observed challenges, and promising directions for future research.

C. Problem Formulation

Given a sentence $S = \{w_1, w_2, \dots, w_n\}$, we define an undirected graph $G = (V, E)$ such that each word $w_i \in S$ corresponds to a node $v_i \in V$. An edge $(v_i, v_j) \in E$ exists if w_i and w_j are syntactically related according to an unknown grammar function. The whole task of the **ROOT NODE IDENTIFICATION** is to learn a function $f : G \rightarrow v^*$ that maps a graph G to a single root node $v^* \in V$, representing the most "important" word in the sentence.

Example of a Syntactic Tree Represented as an Undirected Graph

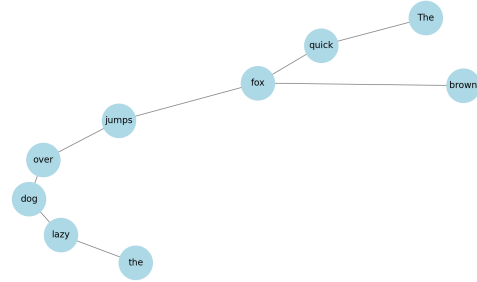


Fig. 1. Example of an english syntactic tree represented as a graph

II. DATA EXPLORATION AND PROCESSING

In this section, we present an overview of the dataset and the preprocessing steps applied to prepare it for modeling. We begin by describing the structure and characteristics of the raw data, followed by the transformations undertaken to convert the sentence-level input into a node-level dataset suitable for supervised learning. We then detail the feature engineering process used to extract structural graph properties for each word. Finally, we analyze the statistical and distributional properties of the resulting features better to understand the dataset's complexity and modeling challenges.

A. Dataset Description

The raw data consists of two sets: 10,500 rows for training and 10,395 for testing, representing a total of 20,895 sentences across 21 languages (995 unique sentences per language).

Each row corresponds to a sentence and includes the following attributes:

- **id**: Unique row identifier.

- **language:** Language of the sentence.
- **sentence:** Sentence ID (1–995, repeated across languages).
- **n:** Number of words (nodes) in the sentence.
- **edgelist:** Pairs of nodes indicating semantic relationships between words.

B. Preprocessing

The first step in preprocessing ensures that the model receives only the intended information. Specifically, we want the learning algorithm to model the relationship between undirected graph measures and whether a word is the root of a sentence.

Given that the data is derived from real sentences, we might infer that the edgelist originally represents a directed graph. However, to prevent the model from learning unintended patterns, such as edge directionality, node ordering, or in-degree features (which may introduce data leakage), we take the following precautions:

- 1) **Treat the edgelist as an undirected graph:** We discard any directionality, avoiding features like in-degree or out-degree.
- 2) **Shuffle edge pairs randomly:** This breaks any implicit ordering derived from the sentence structure.
- 3) **Permute nodes within each edge:** For each edge pair, we randomly swap the nodes to eliminate directional cues further.

Next, to formulate the ROOT NODE IDENTIFICATION task as a binary classification problem, we converted the dataset from a sentence-level to a word-level format, where each row now represents a word (node) labeled as root or not.

This transformation required iterating through the original data to extract word-level features. To optimize the process, we computed some features during the same iteration.

The resulting dataset includes:

- **language:** Language of the original sentence.
- **sentence:** Sentence ID.
- **n:** Number of nodes.
- **node:** Word (node) identifier.
- **is_root:** Label pointing to whether the node is the root in the original sentence (1) or not (0). Only in the train set.
- **Centrality features:** Several metrics used as model inputs (see Section II-C).

Note that since sentence IDs range from 1 to 995 in every language, a sentence is uniquely identified by the (**language**, **sentence**) pair, so we need to keep this information.

C. Feature Engineering

We engineered 20 graph-based features per node (word) using `networkx`, aiming to characterize its structural role in the sentence’s semantic graph. These features fall into four categories:

- **Degree-based Features:** Quantify direct connectivity.
 - degree, degree_squared: Node connectivity and its amplified effect.

- degree centrality: Normalized degree.
- is_leaf, is_hub: Binary indicators for peripheral or highly connected nodes.

- **Neighbor-informed Features:** Capture context from adjacent nodes.
 - degree_diff, local_degree_ratio: Compare node degree with neighbors’.
 - max_neighbor_degree, avg_neighbor_deg: Capture neighborhood connectivity.
- **Centrality Measures:** Reflect importance in the overall graph.
 - harmonic centrality, closeness centrality: Based on shortest path distances.
 - betweenness centrality, pagerank: Capture flow and influence.
 - eccentricity, avg_shortest_path_length: Distance to other nodes.
- **Topological Features:** Describe structural complexity.
 - clustering, triangle_count: Measure local cohesion.
 - kcore_number: Node participation in dense sub-graphs.
 - local_entropy, global_entropy: Degree distribution diversity locally and globally.

Further analysis of each of the features is provided in the notebook `preprocessing.ipynb` [3].

D. Exploratory Data Analysis

The most essential characteristic of the dataset is its significant **class imbalance**: over 90% of the nodes are labeled as non-root. This imbalance naturally arises from the structure of the data, where each row represents a word (node), but only one root exists per sentence. This imbalance poses a challenge for modeling, and appropriate techniques will be applied to account for it during model development.

Moreover, from the raw descriptive statistics, we observe that sentence length varies significantly across the dataset, with a mean of approximately 11.7 words per sentence and a range spanning from 3 to 70 words. This wide variability introduces additional complexity in modeling, as evidenced by the standard deviation of 8.3, indicating a substantial spread around the mean. Similarly, the number of nodes per sentence graph ranges from 1 to 70, with an average of roughly 22 nodes and considerable dispersion. Degree-related metrics, such as degree, degree squared, and degree difference, exhibit a wide range of values, with maximums exceeding 10 in some cases, indicating the presence of highly connected nodes that can influence graph dynamics.

Interestingly, more complex measures, such as local entropy and global entropy, have means near zero with large standard deviations (close to 1), reflecting heterogeneous information content across sentence graphs. Furthermore, centrality measures such as eccentricity and closeness centrality also vary considerably, pointing to diverse node importance

profiles within the graph structures.

Notably, features traditionally expected to contribute discriminative power, such as the clustering coefficient and triangle count, were found to carry little to no discriminative information in this setting and were therefore excluded from further analysis, pending a deeper investigation into their unexpectedly poor performance. More information can be found inside the notebook `preprocessing.ipynb`.

a) Correlation Analysis and Principal Component Analysis: An inspection of the correlation matrix 2 reveals that the degree variable is among the most informative, albeit highly correlated with several other centrality scores. For instance, pagerank and degree centrality exhibit nearly perfect positive correlation, suggesting redundancy among these features. Furthermore, sentence_length exhibits strong negative correlations with features such as global_entropy and kcore_number, likely as a consequence of normalization procedures applied during preprocessing.

Additionally, the three principal components extracted via PCA show distinct relationships with the original variables. Degree-based centrality measures primarily influence the first component, while the second component aligns more with topological descriptors such as is_leaf, avg_neighbor_degree, and avg_shortest_path_length. The third component appears to carry less explanatory power, although it is moderately associated with eccentricity and avg_shortest_path_length.

b) Distributional Properties: The distributions of the numerical features 3 reveal several informative patterns. Sentence_length follows a near-Gaussian distribution centered around 20 tokens. In contrast, degree-related and centrality measures tend to follow skewed, exponential-like distributions, reflecting the scale-free nature often observed in syntactic graph structures. Other variables such as max_neighbor_degree, avg_neighbor_degree, eccentricity, global_entropy, and avg_shortest_path_length exhibit gamma-like distributions, suggesting heterogeneity in local neighborhood structures and path-based metrics across different graphs.

III. ROBUST AND REPRODUCIBLE MODEL DEVELOPMENT

We fixed the random seed to 42 across all experiments, ensuring reproducible results. Moreover, note that the complete source code is publicly available in the corresponding repository [3].

Several critical aspects were considered during model development, independent of the specific algorithm employed:

- **Prediction Strategy:** To avoid the possibility of predicting multiple root nodes per sentence, we rely on probabilistic models rather than binary classifiers. For each sentence, the node predicted as the root is the one with the highest probability of belonging to the positive class (i.e., being the root). This decision rule is formally expressed as:

$$\hat{root} = \arg \max_{v \in V} P(y_v = 1) \quad (1)$$

where V denotes the set of nodes in a given sentence and $P(y_v = 1)$ is the predicted probability of node v being the root.

However, in cases where multiple nodes receive the same maximum probability, often due to uniformly low values arising from the strong class imbalance, a tie-breaking heuristic must be applied. Popular implementations select the first node among the tied candidates, resulting in many nodes being predicted as the root simply because they appear first in the list. By examining the data, we observed that root nodes often occur near the beginning of sentences. This can lead to artificially inflated performance, as many root predictions may be correct merely because the root node frequently appears early in the sentence. Consequently, the model may appear effective, but it may actually rely on positional bias rather than learned structural patterns. To mitigate this issue, we randomized node order during preprocessing, as described in Section II-B, thereby eliminating position-based advantages.

- **Hyperparameter Optimization:** During hyperparameter tuning it is essential to ensure that all nodes belonging to the same sentence remain in the same fold. This constraint preserves the structural coherence of the data, as sentences are the natural unit of observation. Failing to do so would result in information leakage between training and validation sets.
- **Class Imbalance:** The dataset is heavily imbalanced, with over 90% of nodes labeled as non-root. This skews model predictions toward the majority class, reducing effectiveness on the minority (root) class—our main focus. To mitigate this, we applied strategies from the imbalanced-learn (imblearn) library [4]:

- **Resampling:** *Oversampling* (e.g., SMOTE [5], ADASYN [6]) increases minority instances; *undersampling* (e.g., NearMiss [7], TomekLinks [8]) reduces the majority class; hybrid approaches like SMOTEENN and SMOTETomek [9] combine both.
- **Cost-Sensitive Learning:** Assigning higher misclassification costs to the minority class via *class weights* (e.g., in SVMs [10], tree-based models) or *custom loss functions* (e.g., focal loss [11]) improves sensitivity to root nodes [12], [13].
- **Imbalance-Aware Ensembles:** Ensemble methods tailored for imbalance, such as BalancedRandomForestClassifier, BalancedBaggingClassifier, and EasyEnsembleClassifier [14], [15], are available in imblearn.
- **Threshold Adjustment:** Post-processing by tuning probability thresholds—based on F1-score or precision-recall curves [16], [17]—can improve minority class detection. While our model selects the highest-probability node, understanding probability calibration remains important.

IV. RESULTS

This section summarizes the most significant modeling outcomes from our experiments. We distinguish between *unilingual classifiers*, which train separate models for each of the 21 languages, and *multilingual classifiers*, which train a single model incorporating language as an explicit feature. All reported accuracies are sentence-level classification accuracies on the test set, meaning the percentage of sentences for which the root node was correctly identified.

- **Logistic Regression (Multilingual):** As a simple baseline, a multilingual Logistic Regression model was trained. This model, despite its simplicity and without explicit handling of class imbalance beyond what the solver might implicitly do, achieved a sentence-level accuracy of **23%** on the test set. This serves as a basic benchmark for more complex approaches.
- **Multilingual Random Forest:** A single Random Forest model was trained on the entire dataset, incorporating language as a feature. To address the pronounced class imbalance, we utilized the `class_weight='balanced'` hyperparameter available in scikit-learn’s implementation. This automatically adjusts weights inversely proportional to class frequencies. Cross-validation was performed ensuring that all nodes from the same sentence remained in the same fold. This model resulted in a **29%** sentence-level accuracy. The improvement over Logistic Regression highlights the Random Forest’s ability to capture more complex non-linear patterns and the benefits of its bagging approach with decision trees.
- **Unilingual LightGBM:** An ensemble of 21 LightGBM models was trained, one for each language. Each model was cross-validated independently. Without extensive hyperparameter tuning beyond default settings (or minimal adjustments), this unilingual approach achieved a sentence-level accuracy of **26%**. This suggests that language-specific patterns might be beneficial, though perhaps not as strongly captured as with the multilingual Random Forest in this instance without further tuning.
- **Unilingual Support Vector Machines (SVM):** We experimented with Support Vector Classifiers (SVC) in a unilingual setting. Using default hyperparameters, the ensemble of SVCs yielded an 7% sentence-level classification rate. After performing a grid search for hyperparameter optimization (e.g., for C , γ , and kernel type) for each language-specific model, this rate improved to **9%**. The decision to pursue a unilingual strategy for SVMs was heavily influenced by their computational complexity, which can be up to $\mathcal{O}(N^2)$ or $\mathcal{O}(N^3)$ for training with N samples, depending on the implementation and kernel. Training 21 separate models, each on approximately $1/21^{st}$ of the data, proved to be computationally more tractable than training a single, large multilingual SVM, despite the overhead of managing multiple models.
- **Unilingual Random Forest:** This approach involved training a separate Random Forest model for each of the 21 languages. For each language-specific model, class imbalance was addressed by computing the specific class imbalance for that model. K-fold cross-validation was carefully implemented, ensuring that all nodes belonging to the same sentence were always kept in the same fold to prevent data leakage and maintain the integrity of sentence-level predictions. The predictions from these 21 models were then concatenated to form the final test set predictions. This strategy yielded the best overall performance, achieving a sentence-level accuracy of **31%**. The superior performance of the unilingual Random Forest ensemble can be attributed to several factors:
 - **Automated machine learning:** To further investigate different scikit-learn models specifically, we also used an automated machine learning (AutoML) pipeline, eZAutoML, and explored various modeling strategies and hyperparameters. Two notable discoveries include:
 - **HistGradientBoostingClassifier (Unilingual):** When deployed in a unilingual ensemble (one model per language), the HistGradientBoostingClassifier, known for its speed and efficiency with large datasets, achieved a competitive sentence-level accuracy of just over **30%**. This performance is comparable to the best-performing Unilingual Random Forest, highlighting its potential.
 - **AdaBoostClassifier (Unilingual):** An ensemble of unilingual AdaBoostClassifiers, leveraging decision trees as base estimators, achieved a sentence-level accuracy of **25%**. While not outperforming the top models, AdaBoost demonstrated its ability to construct a strong classifier, nevertheless these results slightly disappointing.
- **CatBoost:** After evaluating various popular gradient boosting methods, we chose to experiment with CatBoost, given its well-known advantages in handling categorical variables. We conducted various multilingual model setups, and discovered an interesting limitation: CatBoost generates temporary files in a local `catboost_info` directory, which is not shared across processes. This makes it difficult to train unilingual models in parallel without delving into the tool’s internal design, therefore making them unfeasible to train. Given the simplicity and efficiency of the multilingual setup, we opted to apply advanced Bayesian optimization frameworks such as Optuna to tune hyperparameters according to our custom evaluation metric. This approach led to an impressive 30% improvement—currently the best performance achieved by any multilingual model to date.
- **Multi-Layer Perceptron (MLP):** Lastly, we addressed the usage of an unilingual ensemble of MLP’s, from the initial hypothesis that each language must have its own pattern of syntactic rooting. By training a regularized, ADAM-optimized fairly large (6 hidden layers) neural network, we achieved an overall correct sentence classification score of 24% in the test set. Despite the efforts on overfitting this model to then hand tune regularization, MLP has reached a local optima, more nuanced neural networks need to be evaluated in the future.

V. DEFINITIVE MODEL SELECTION AND PERFORMANCE SUMMARY

After extensive experimentation with modeling approaches, feature sets, and imbalance strategies, the **Unilingual Random Forest ensemble** emerged as the best-performing model for root node identification, achieving a sentence-level accuracy of **31%** on the held-out test set.

Its superior performance stems from several key factors:

- **Language-Specific Adaptation:** Training separate Random Forests for each of the 21 languages enabled the ensemble to capture language-specific syntactic and semantic graph patterns, avoiding the generalization limitations of a single multilingual model.
- **Non-linearity and Robustness:** Random Forests model complex, non-linear relationships and are less prone to overfitting than individual trees—crucial for handling graph-structured data.
- **Localized Imbalance Handling:** Using `class_weight='balanced_subsample'` within each model allowed for language-specific adjustment to class imbalance, outperforming global strategies.
- **Ensemble Aggregation:** Combining predictions from specialized unilingual models leveraged diverse linguistic expertise, improving performance on the multilingual test set.
- **Reliable Cross-Validation:** Ensuring sentence-level grouping during cross-validation prevented data leakage and improved generalization.

TABLE I
SUMMARY OF MODEL PERFORMANCE FOR SENTENCE-LEVEL ROOT
NODE CLASSIFICATION

Model Architecture	Strategy	Test Set Accuracy (%)
Support Vector Machine (tuned)	Unilingual	9.0
Logistic Regression	Multilingual	23.0
Multi-Layer Perceptron	Unilingual	24.0
AdaBoost Classifier	Unilingual	25.0
LightGBM	Unilingual	26.0
Random Forest	Multilingual	29.0
CatBoost	Multilingual	30.0
HistGradientBoosting	Unilingual	30.2
Random Forest	Unilingual	31.0

VI. CONCLUSIONS

This investigation into Root Node Identification ended with a Unilingual Random Forest ensemble achieving a 31% sentence-level accuracy. While this represents a significant improvement over simpler baselines and demonstrates that graph-theoretic features contain a discernible signal, it also underscores the inherent difficulty of this task with the available information. A key learning from this project is the profound challenge posed by the abstract nature of the dataset and the limitations encountered when relying solely on the provided graph topology and our engineered structural features. Despite developing 20 distinct centrality and topological measures, and observing their individual correlations (e.g., ‘degree_diff’, ‘closeness_centrality’), pushing performance significantly beyond the 30% ceiling proved exceptionally difficult. This

suggests that while these structural properties are relevant, they may not be sufficiently discriminative on their own to unambiguously identify the “root” word, especially given its underspecified definition in the problem statement.

The journey itself provided critical insights into the modeling process:

- **Addressing Class Imbalance:** The severe skew (over 90% non-root nodes) was a dominant factor. Our experiments confirmed that generic approaches are insufficient, and tailored strategies, such as the per-language class weighting within the Unilingual Random Forest, were crucial for any meaningful progress.
- **Meticulous Data Preprocessing:** Steps like randomizing node and edge order were vital to prevent models from exploiting superficial positional biases rather than learning genuine structural patterns. This reinforced the need to understand potential data leakage points.
- **Systematic and Justified Model Exploration:** Moving from simple baselines (Logistic Regression) to more complex ensembles (Random Forests, Gradient Boosting variants) was not just about trying different algorithms, but understanding *why* certain models might be better suited. For instance, the computational cost of SVMs steered us towards a unilingual strategy, while the robustness of Random Forests to feature interactions and their inherent imbalance handling options made them a strong candidate.
- **The Value of Unilingual Approaches:** For this dataset, allowing models to specialize on language-specific patterns yielded consistently better results than attempts to create a single, all-encompassing multilingual model, highlighting that “rootness” cues may indeed vary linguistically.

Further significant improvements would likely require exploring avenues beyond the current feature space and modeling paradigms. For instance, incorporating explicit linguistic features could have provided richer, more discriminative signals. Similarly, applying Graph Neural Networks (GNNs) might have enabled the learning of more nuanced representations directly from the graph, potentially overcoming the limitations of handcrafted features. More sophisticated feature selection or advanced ensemble techniques could also have been explored. For the exploration of further multilingual models, Optuna and other Bayesian optimization or RL based methods for hyperparameter tuning are greatly recommended.

Overall, this project served as a comprehensive exercise in applied machine learning, emphasizing that while sophisticated models are powerful, their success is deeply connected with the quality and relevance of input data, a methodical approach to experimental design, and a clear understanding of the problem’s uncommon and inherent complexities and limitations.

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APPENDIX A IMAGES

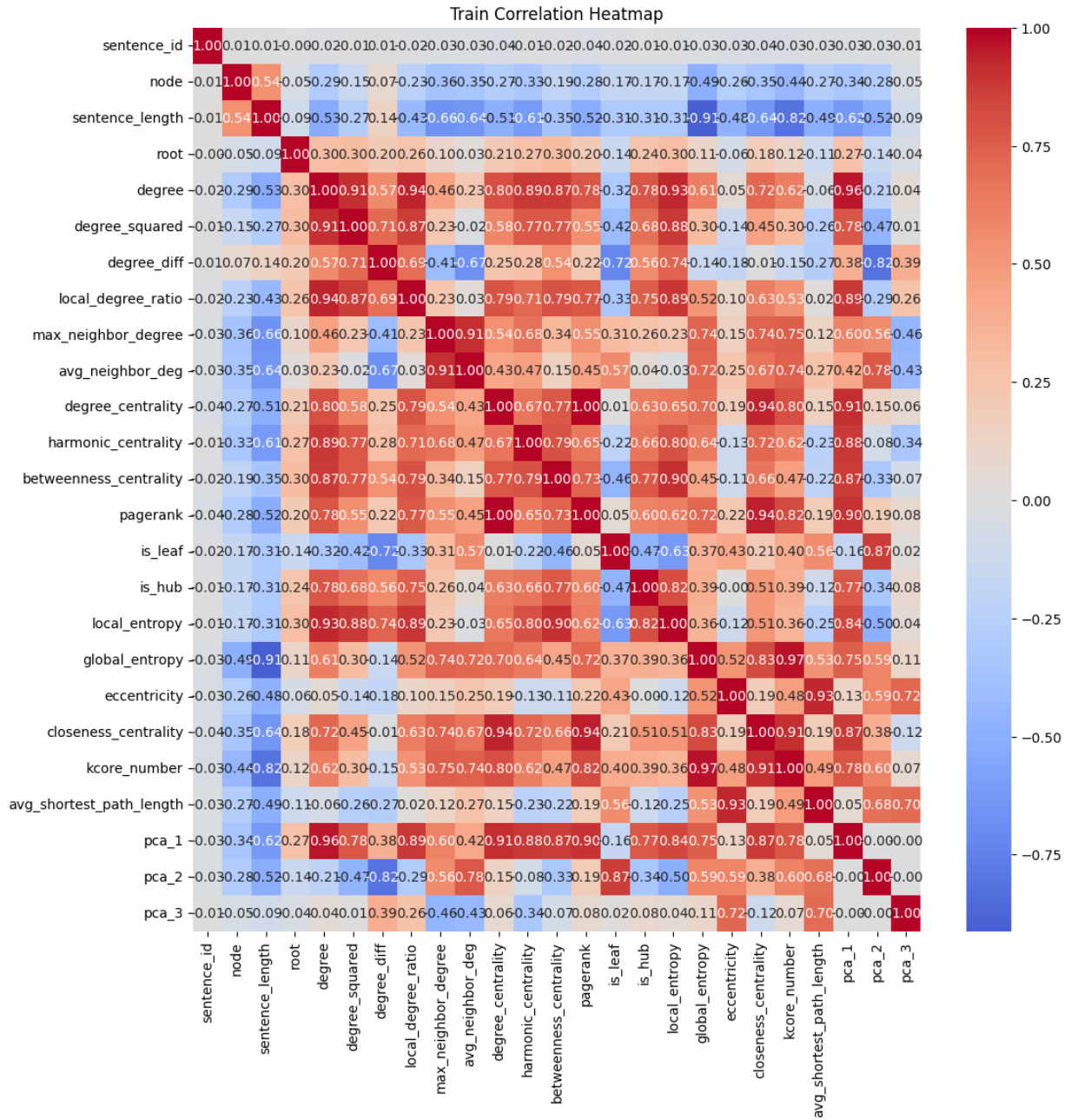
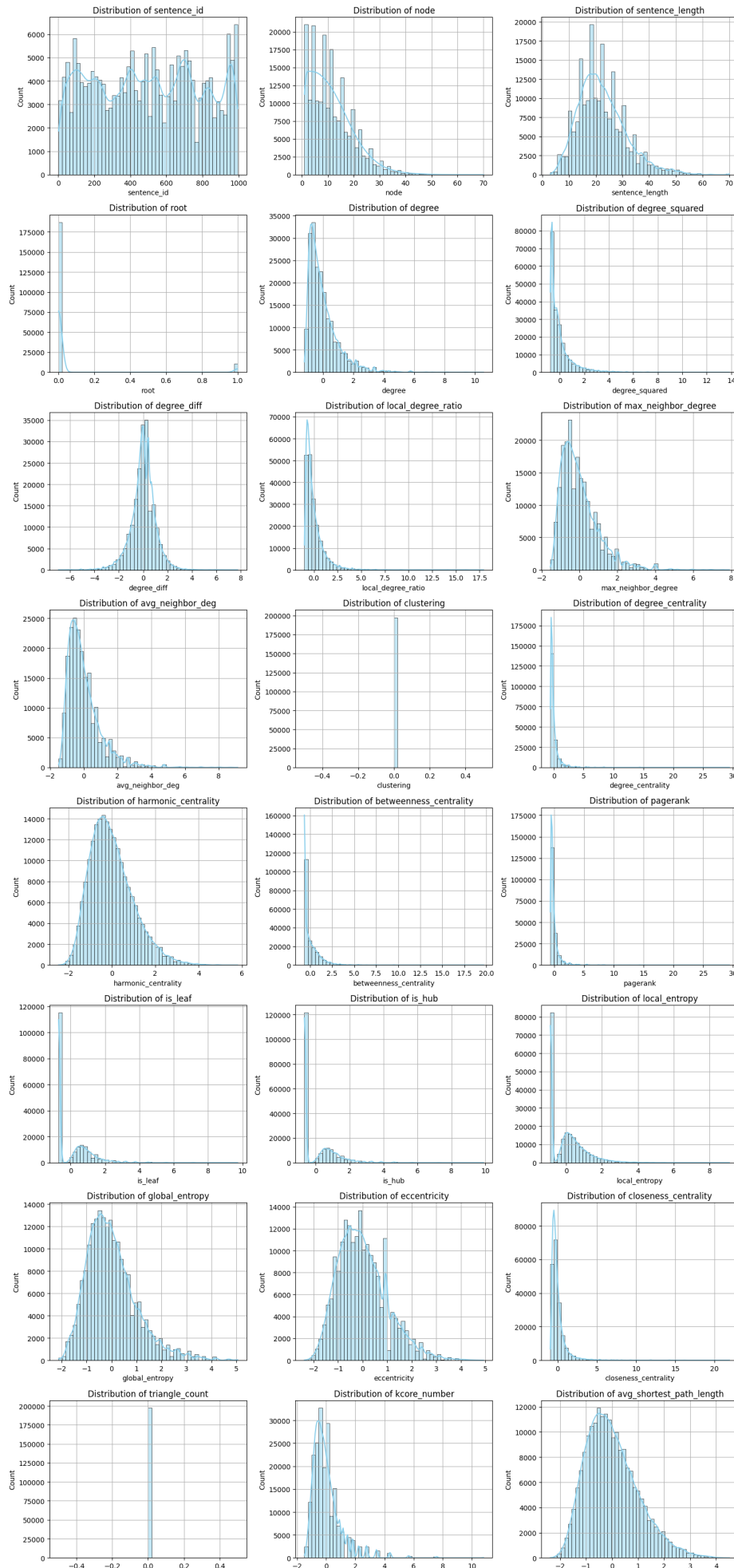


Fig. 2. Correlation heatmap of all (X, y) variables used throughout the project



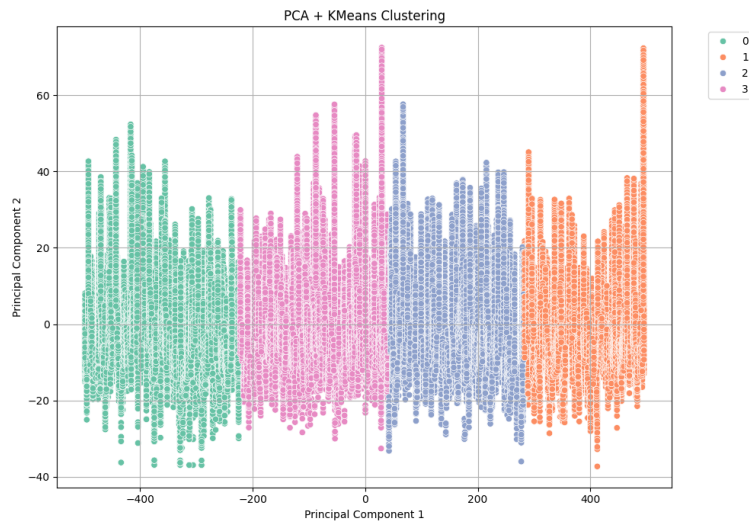


Fig. 4. Results of clustering applied to principal component analysis dimensionality reduction by using 2 components

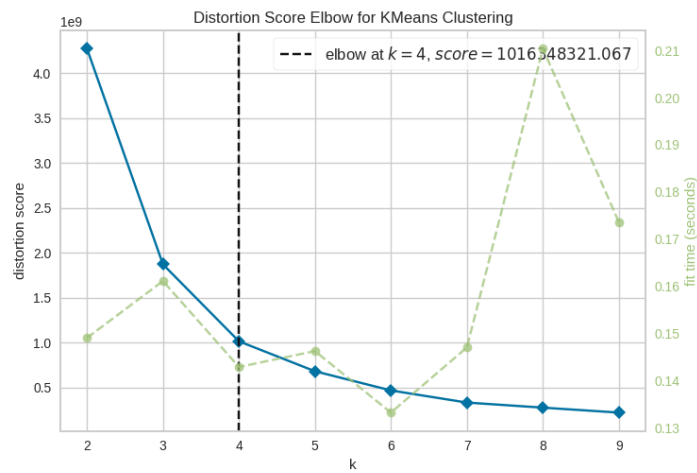


Fig. 5. Distortion score as a function of the number of clusters in the k-means algorithm. According to the empirical elbow method, 4 clusters provide the best trade-off between model complexity and fit quality.